

SAFETY QUIZ:CHE382-Chemical Engineering Laboratory II

Please answer the following questions testing your knowledge from 6 selected chapters of the JT Baker Hazardous Chemical Safety document; Toxic Chemicals, Flammables, Corrosive Chemicals, Reactive Chemicals, Insidious Hazards and Compressed Gases. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words. (5 points per question)

1. What is the general definition of a toxic chemical?

A toxic material is any chemical in relatively small amounts when ingested, inhaled, or absorbed causes damage to structure or function.

2. Name the general types of damage that a toxic chemical can cause:

1. Damage to structure
2. Disturbance to Function



3. Name two parameters that can effect toxicity:

Two parameters that can affect toxicity are local the duration of the exposure and the dose of the exposure.

4. What is the DOT (Department of Transportation) definition of a nonflammable compressed gas?

The DOT defined nonflammable compressed gas as having a psia greater than 40 at 70F or a psia greater than 104 at 130F or psia greater than 40 at 100F

5. a) How does one determine how much material is remaining in a non-liquified gas cylinder?



The only way to determine how much liquified gas remains in a cylinder would be to weight it and compare it to the net weight of the contents provided on the cylinder.

- b) On page 13-10, which pressure gauge on the two-stage regulator will give you the cylinder pressure reading?



The cylinder pressure reading can be found the right most pressure gauge.

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6. Other than a single compressed gas, what else can be stored inside a compressed gas cylinder?



Inside a compressed gas cylinder, one could also store liquids.

7. Why do insidious hazards represent a problem to workers?

Insidious hazards represent a problem to workers because they are not usually seen, smelled, tasted, or felt, and can be overlooked or ignored.

8. What are the four general categories of insidious hazards?

The four general categories of insidious hazards are:

1. Unanticipated Chemical Reactions
2. Unanticipated Toxicity
3. Improper Facility Design
4. Improper Handling of Equipment and Chemicals

9. Name 2 control measures that can be implemented to reduce insidious hazards in the laboratory.

Two control measures that can be implemented to reduce insidious hazards in the lab are to have suitable disaster equipment available and to not allow stock build-up of toxic, flammable, or corrosive materials.

10. What is the definition of a reactive chemical?

A reactive chemical is a substance which enters a violent reaction during spontaneous liberation of heat and/or gases too rapidly.

11. Name the two groups and their sub-headings used to categorize reactive chemicals:

Two categories of reactive chemicals are unstable compounds such as explosive and monomers and then unstable mixtures such as water-reactive compounds or air sensitive compounds

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12. List two precautionary measures for handling and/or storing reactive chemicals.

Two precautionary measures for handling and storing reactive chemicals are to store a minimum quantity of material and to store in isolated, heavy constructed, preferable underground walls.

13. How are corrosive chemicals generally defined?

Corrosive chemicals are substances that injure body tissue or are corrosive to metal by direct action

14. How do corrosive chemicals act on body tissues?

Corrosive chemicals can dissolve skin fats, soften the epidermis, and sensitize skin to other chemicals.

15. True or False: Concentrated alkaline solutions have a more corrosive effect on tissue than most strong inorganic acids.

True

16. Name the six categories of common corrosive liquids.

The six categories of common corrosive liquids are Inorganic Acids, Organic Acids, Other Inorganics, Caustic Solutions, Organic Solvents, and Other Organics.

17. What category of corrosive solids presents the greatest hazard because of their frequent use in industry?

The category of corrosive solids that present the greatest hazards because of frequency of use is "dust"

18. Name the three components that make up the FIRE TRIANGLE:

The three components of a fire triangle are the oxidizer, the fuel and the energy.

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19. What category of flammables is the most common source of fires in industry?

The category of flammables that serve as the most common source of fires in industry is dust.

20. List two common safety problems and two safety control measures for flammable solvents.

Two common safety problems for flammable liquids are no floor level ventilation and improperly stored glass containers for the flammable solvents. To control these two problems one can, eliminate sources of ignition and isolate solvents from reactive chemicals.

BONUS QUESTION:

Where is the eyeglass cleaning station located in the Unit Operations Laboratory (Room 120)?



The eyeglass cleaning station is located at both of the entering doors.

Comment Summary

Page 1

1. -2
Local toxicity, systemic toxicity, acute toxicity, chronic toxicity
2. the question asked for NON-liquified gas
3. -2.5
Read the cylinder pressure gauge reading which is proportional to the amount of gas remaining inside the cylinder
4. -2.5
the left one

Page 2

5. -5
Liquefied gas, a gas dissolved in a liquid or solid, or a compressed gas mixture

Page 3

6. They act through direct contact with skin or eyes, inhalation or ingestion, and by direct chemical reaction, dissolution of essential components, destruction of protein or disruption of cell membranes.
7. -5
Caustic alkalies

Page 4

8. only at the west entrance